Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-uniqueness of the moment problem for Schwartz functions

Files.

- File: `briefs/SchwartzMomentProblem/SchwartzMomentProblem.lean`
- Library: `SchwartzMomentProblem`
- Imports: `Mathlib.Analysis.Calculus.BumpFunction.InnerProduct`, `Mathlib.Analysis.Distribution.SchwartzSpace.Basic`, `Mathlib.MeasureTheory.Measure.Lebesgue.Basic`

Setting. Schwartz functions $f, g \in \mathcal{S}(\mathbb{R}, \mathbb{C})$ (`SchwartzMap`) with compact supports and $\operatorname{tsupp} f \cap \operatorname{tsupp} g = \emptyset$, and $\psi_\theta = f + e^{i\theta} g$. The moments are

$$\langle \psi | X^m P^n | \psi \rangle = \int_{\mathbb{R}} \overline{\psi(x)}\, x^m\, (-i\hbar)^n\, \psi^{(n)}(x)\, dx,$$

with $\hbar$ an arbitrary real parameter.

Results.

Brief	Lean
cross terms vanish, pointwise eq. (3): $\langle\psi_\theta\|X^m P^n\|\psi_\theta\rangle = \langle f\|X^m P^n\|f\rangle + \langle g\|X^m P^n\|g\rangle$	`conj_mul_iteratedDeriv_eq_zero`, `density_psi` `moment_psi`
each θ gives a different state	`linearIndependent_psi`: $\psi_\theta, \psi_{\theta'}$ linearly independent when $e^{i\theta} \neq e^{i\theta'}$
the bump functions of Fig. 1	`bump`, `tsupport_bump_subset`, `bump_ne_zero`
Theorem 1	`exists_same_moments`, with $\psi_1 = \psi_0$, $\psi_2 = \psi_\pi$ for bumps centred at 0 and 3

Differences from the brief.

- The moment is defined as an integral for every Schwartz function, but the identity for ψ_θ is proved only under the brief's hypothesis that f and g have compact support. This makes every integrand continuous with compact support, hence integrable.
- "Each θ defines a different physical state" is formalized as linear independence of ψ_θ and $\psi_{\theta'}$, which holds exactly when $e^{i\theta} \neq e^{i\theta'}$, i.e. for $\theta \not\equiv \theta' \pmod{2\pi}$.
- The closing remark on Schwartz seminorms and the conclusion about local operators are not formalized.

Problems found in the brief. No mathematical problems. The argument needs f and g to be nonzero, which the brief leaves implicit, and "different state for each θ" holds only modulo 2π.